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Report

Racial and ethnic disparities in breast cancer rates by age: NAACCR Breast Cancer Project

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Summary

Objective. To examine age-specific rates of breast cancer incidence among racial and ethnic groups in the United States.

Methods. Subjects were 363,801 women diagnosed with invasive breast cancer diagnosed during 1994–1998 and reported in the North American Association of Central Cancer Registries (NAACCR) data set. Variables analyzed included race, ethnicity, 5-year age group (from 10 years through 85+ years), and stage at time of diagnosis (localized, regional, distant). Incidence rates per 100,000 women were calculated for each 5-year age group and stratified by stage. Rate ratios and 95% confidence intervals were calculated by comparing each racial group with whites and Hispanics with non-Hispanics.

Results. Black women experience significantly higher breast cancer incidence up to the age of 40 years and significantly lower incidence after age 50 compared with white women of the same ages. This is called the 'crossover' effect. This shifting burden of higher incidence occurs at ages 35–39 for localized stage and at ages 55–59 for regional stage. For distant stage, black women of all ages experience higher incidence compared with white women. Similar crossover effects do not exist for American Indian (AI) or Asian/Pacific Islander (API) women compared with white women. Both AI and API women have significantly lower incidence of breast cancer compared with white women, and Hispanic women have significantly lower incidence compared with non-Hispanic women.

Conclusions. This study highlights racial and ethnic differences in breast cancer incidence rates among US women. The crossover effect between black and white women, particularly the lower incidence of localized stage disease diagnosed in older black women, is a significant phenomenon that may be associated with screening practices, and has implications for public health planning and cancer control initiatives to reduce racial/ethnic disparities.

Introduction

Breast cancer is the most commonly diagnosed invasive cancer in the United States for women of all racial and ethnic groups, with an estimated 211,300 new cases diagnosed in 2003 [1]. Two key characteristics of women that significantly affect breast cancer incidence are age and race. Breast cancer diagnosed in younger women is often more aggressive [2]. In addition, unlike countries with low rates of breast cancer incidence, US rates continue to rise after menopause and are highest in the older age categories. With respect to racial patterns, overall age-standardized rates are higher among white women than black women [1]. Race and age interact in their effect on breast cancer incidence, and a 'crossover' effect between black and white women has been shown in previous studies [3]. This effect manifests as higher rates among younger black women and lower ones among older black women, relative to white women of the same age [3]. Incidence rates for Asian/Pacific Islander, American Indian/Alaska Native, and Hispanic women are generally lower than those for US white or black women, although previous reports have been based on limited populations and small samples [4].

Although researchers have examined differences in breast cancer incidence between black and white women [4–6], less is known about differences among other racial and ethnic groups. In addition, the crossover effect has not been documented in recent data. This paper focuses on age-specific rates of breast cancer incidence among racial and ethnic groups, with a more detailed and up-to-date examination of age-related patterns.

Materials and methods

Source of data: the NAACCR Breast Cancer Research Data Set

The North American Association of Central Cancer Registries (NAACCR) Breast Cancer Research Data Set covers the 5-year period from 1994 through 1998. It includes data from state and city cancer registries that meet the NAACCR standards for high data quality [4]. First, a registry must have submitted data for each of the 5 years included in this study, 1994–1998. Second, registries were excluded if they had one or more duplicate reports for an individual cancer occurrence for one person per 1000 records in the registry data file. Third, all files were evaluated for coding accuracy and reliability using a standard editing program; registries included in the data set are those found to be error-free for variables used in the computation of cancer incidence rates. Finally, all registries included in the data set had 90% or higher case ascertainment rates for each of the 5 years, after adjusting for duplicate reports [8]. Of the registries meeting these standards, the following consented for their data to be used in this study: Arizona, Colorado, Connecticut, Delaware, Metropolitan Atlanta (Georgia), Hawaii, Idaho, Illinois, Iowa, Kentucky, Louisiana, Metropolitan Detroit (Michigan), Montana, Nebraska, New Hampshire, New Jersey, North Carolina, Pennsylvania, Rhode Island, Utah, Washington, West Virginia, Wisconsin, Wyoming, Los Angeles (California), and San Francisco Bay Area (California). These registries cover approximately 40% of the US population.

To determine if a person is Hispanic, most US registries use hospital records. Computer algorithms may be used, but never as the sole source for determining ethnicity. Population data on Hispanics are often available for large areas, but seldom by county. NAACCR coding guidelines for Hispanic ethnicity are used most often, but surname and place of birth may be employed. Maiden name is often collected, and about half of the registries use this information to determine Hispanic ethnicity [8]. During the time of this study, NAACCR had no national standard guidelines for determining Hispanic ethnicity. Lack of comparable definitions of Hispanic ethnicity across registries may have resulted in inaccurate analyses, and results of this study are interpreted conservatively.

Population counts, used as denominators for calculating cancer incidence rates, were derived from post-1990 US census estimates [9]. Population estimates for Asian/Pacific Islanders and whites living in Hawaii were adjusted for an assumed undercount of non-white populations.

Ethnic and racial classification

The four standard US census race classifications with available denominator data are: white, black, American Indian/Alaska Native (AI/AN), and Asian/Pacific Islander (API). Ethnicity is categorized as Hispanic or non-Hispanic across all racial groups. Data specific to race and ethnicity were included in this study only for those registries that reported at least 20 invasive female breast cancer cases and at least 100,000 total women in a particular subgroup. This minimum value of inclusion for the numerator is to ensure confidentiality of cases. The denominator value was chosen to minimize instability of rates in small populations because data collection for minority groups is not standard across all registries. Registries included in the analyses of Hispanic versus non-Hispanic women could not have more than 3% of female breast cancer cases classified as unknown Hispanic ethnicity.

All 26 consenting registries were included in the analyses of white populations. For other racial and ethnic groups, the contributing registries differed and are identified in the footnote for each table. Because rates for AI/AN are based on data from only five states, few, if any, Alaska Natives are included. Thus, the term, 'American Indian' (AI) is used throughout this paper, even though the standard US census category of 'American Indian/Alaska Native' was the source for population counts.

Because of these restrictions, analyses presented in this paper include invasive incident female breast cancer diagnoses occurring among 40% of white, approximately 37% of black, 26% of AI, 52% of A/PI, and 27% of Hispanic populations in the US. No rates are presented for subcategories with 20 or fewer breast cancers. Invasive incident female breast cancer is defined by the *International Classification of Diseases for Oncology* under site code C50 [10].

Description of subjects and variables

A total of 363,801 invasive incident breast cancer cases diagnosed among women during 1994–1998 were included in this data set. Diagnoses for 98.3% of these were microscopically confirmed, and 0.9% (3137) of the total number of cases were based on autopsy findings or death certificate only. If a case was diagnosed by death certificate only, it would be included in the incidence rate but would be categorized as 'missing' stage.

In addition to race and ethnicity, subjects were stratified by age and stage at the time of diagnosis. Age was categorized by 5-year age groups from 10–14 years through 85+ years, to pinpoint the timing of the crossover effect.

Stage of disease was categorized according to SEER summary definitions for localized, regional, and distant stages [11]. Localized disease stage is defined as cancer confined to the site/organ of origin (breast). Regional stage is defined as direct extension of the cancer to adjacent organs/structures or to regional lymph nodes (axillary, transpectoral, internal mammary). Distant stage includes cases with involvement of distant organs or lymph nodes (including supraclavicular, cervical, or contralateral internal mammary), either by direct extension or through discontinuous metastasis.

Statistical analyses

All incidence rates are per 100,000 women. Average annual age-specific incidence rates were calculated using the method of Fay and Feuer [12]. Rate ratios (RR) were calculated by comparing breast cancer incidence rates of white women with each of the other racial categories, and by comparing rates of Hispanic women with non-Hispanic women. Confidence intervals (95% CI) for the RR were estimated using the formula $\exp(\ln(RR) \pm Z_\alpha SD[\ln(RR)])$ [13].

Results

The crossover effect is apparent between black and white women for all disease stages combined. Black women experience significantly higher breast cancer incidence up to the age of 40 years and significantly lower incidence rates after age 50, compared with white women of similar age (Table 1). Similar crossover effects do not exist for AI or API women compared with white women. Both AI women (over age 40) and API women (over age 30) have significantly lower rates of breast cancer incidence compared with white women (Figure 1), and differences in rates between API and white women widen with age. No differences existed in incidence rates whether analyses were limited to registries with 20 or more cases in a specific racial group or not.

Although caution is warranted when interpreting ethnicity data because of selection methods, rates between Hispanic and non-Hispanic women are not statistically significantly different until the age group 30–34 years. After that, rates for Hispanic women are significantly lower than for non-Hispanic women (Table 2), and differences in rates widen with age. The point of divergence in rates by ethnicity begins at a somewhat younger age (30–34 years) compared with that of black and white women (40–44 years).

When further stratifying breast cancer incidence rates by stage at diagnosis in addition to racial group, several interesting patterns emerge (Table 3). Within the localized stage, the crossover effect between black and white women begins approximately at ages 35–39 years. For AI women over the age of 40 and API women over the age of 30, both groups experience significantly lower incidence rates for localized breast cancer compared with white women.

For cases diagnosed at the regional stage, black women ages 54 years and younger have higher rates of breast cancer incidence compared with white women, but in older age groups, no significant differences are noted. For AI women, rates are lower than those for white women, although sufficient data are available only for women ages 45–64 years. For API women over the age of 30, rates are consistently lower than for white women.

Within the distant stage, black women of all ages experience higher rates of breast cancer incidence than white women (statistically significant for all ages over 30 years). For AI women, no data are presented due to low numbers, and for API women, rates are lower than for white women (significantly lower for women over age 55).

When rates for Hispanic/non-Hispanic ethnic groups are compared within disease stages, lower incidence rates are noted for both localized stage (over age 25) and regional stage (over age 30) (Table 4). Rates among Hispanics were almost always lower than those

Table 1. Average annual age-specific breast cancer incidence rates per 100,000 population for race groups in the NAACCR data base, 1994–1998

Age (years)	White[a]	Black[b]			American Indian[c]			Asian/Pacific Islander[d]		
	Rate	Rate	RR[e]	95% CI[f]	Rate	RR	95% CI	Rate	RR	95% CI
20–24	1.20	2.30	1.92	1.42–2.60	–[g]	–	–	–	–	–
25–29	7.81	12.20	1.56	1.38–1.77	–	–	–	5.29	0.68	0.53–0.88
30–34	25.38	33.21	1.31	1.26–1.36	–	–	–	18.54	0.73	0.64–0.83
35–39	58.20	68.50	1.18	1.12–1.24	21.76	0.37	0.25–0.56	47.74	0.82	0.75–0.89
40–44	115.40	117.59	1.02	0.98–1.04	57.93	0.50	0.38–0.65	97.39	0.84	0.79–0.89
45–49	190.70	184.77	0.97	0.93–1.00	114.28	0.60	0.48–0.75	153.32	0.80	0.76–0.84
50–54	253.25	230.87	0.91	0.88–0.94	122.46	0.48	0.38–0.61	187.67	0.74	0.70–0.78
55–59	301.30	269.46	0.89	0.86–0.92	163.23	0.54	0.43–0.68	221.40	0.73	0.69–0.77
60–64	354.80	285.29	0.80	0.77–0.83	201.39	0.57	0.45–0.72	227.97	0.64	0.60–0.68
65–69	414.53	333.60	0.80	0.77–0.83	136.53	0.33	0.24–0.45	229.86	0.55	0.52–0.59
70–74	466.59	366.04	0.78	0.75–0.81	129.25	0.28	0.19–0.41	242.61	0.52	0.49–0.56
75–79	482.97	394.05	0.82	0.79–0.85	137.34	0.28	0.18–0.43	238.21	0.49	0.45–0.53
80–84	465.23	388.29	0.83	0.79–0.87	–	–	–	218.91	0.47	0.42–0.53
85+	395.10	332.72	0.84	0.79–0.89	–	–	–	156.39	0.40	0.34–0.47

[a] Reference group includes data from all consenting US registries.
[b] Includes data from all consenting US registries except Idaho, Montana, New Hampshire, Utah, and Wyoming.
[c] Includes data from Arizona, Montana, North Carolina, Washington, and Wisconsin.
[d] Includes data from Arizona, Los Angeles and Greater Bay Area (California), Colorado, Connecticut, Metro Atlanta (Georgia), Hawaii, Illinois, Iowa, Kentucky, Louisiana, Metro Detroit (Michigan), New Jersey, North Carolina, Rhode Island, Utah, Washington, and Wisconsin.
[e] Rate ratio comparing each race group rate with that of whites.
[f] Confidence interval.
[g] Rate could not be calculated because fewer than 20 cases were reported.

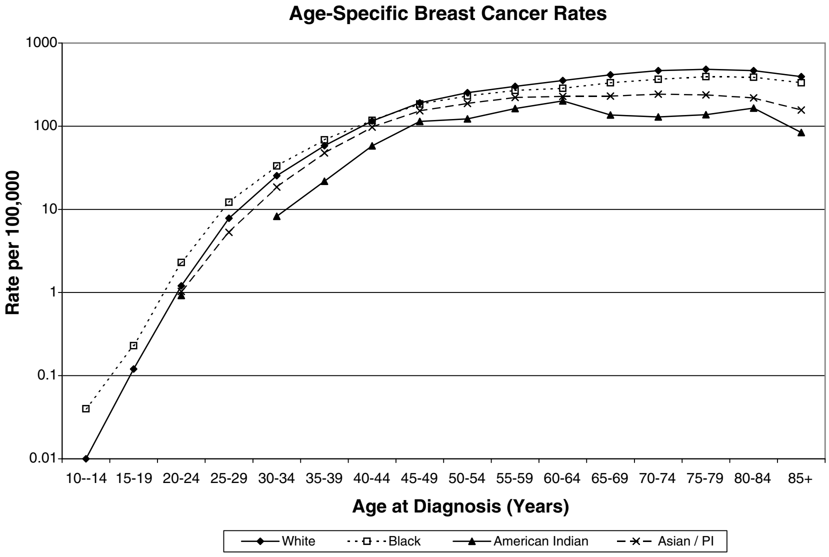

Figure 1. Breast cancer incidence by age and race, NAACCR 1994–1998.

Table 2. Average annual age-specific breast cancer incidence rates per 100,000 population for Hispanic/non-Hispanic[a] ethnic groups in the NAACCR data base, 1994–1998

Age (years)	Non-Hispanic[b] Rate	Hispanic		
		Rate	RR[c]	95% CI[d]
20–24	1.13	1.23	1.09	0.73–1.63
25–29	7.33	6.45	0.88	0.74–1.04
30–34	23.11	17.14	0.74	0.67–0.82
35–39	52.33	36.79	0.70	0.65–0.75
40–44	100.48	73.99	0.84	0.70–0.78
45–49	165.64	118.07	0.71	0.67–0.75
50–54	218.72	145.14	0.66	0.63–0.70
55–59	258.29	169.11	0.65	0.61–0.69
60–64	294.97	198.09	0.67	0.63–0.71
65–69	337.19	219.35	0.65	0.61–0.69
70–74	377.25	231.57	0.61	0.57–0.65
75–79	390.33	232.25	0.59	0.55–0.64
80–84	376.47	210.54	0.56	0.51–0.62
85+	321.75	177.62	0.55	0.49–0.62

[a] Includes data from Los Angeles and Greater Bay Area (California), Colorado, Connecticut, Idaho, Iowa, Kentucky, Metro Detroit (Michigan), Kentucky, Nebraska, Rhode Island, Utah, Washington, and Wyoming.
[b] Reference group for rate ratios.
[c] Rate ratio comparing Hispanic with non-Hispanic rates.
[d] Confidence interval.

for non-Hispanics, although differences were not always statistically significant.

Discussion

Results of this study reveal significant differences in breast cancer incidence rates between racial and ethnic groups, with clear evidence of age patterns. Incidence rates for AI and API women are clearly lower than for white and black women at all ages. Black women have significantly higher incidence rates than white women at ages younger than 40 years, with the largest gap occurring in the youngest (20–29 years) age group. This pattern reverses after age 50, with white women exhibiting significantly higher rates. This crossover effect for white and black women has been documented in the United States since at least the 1960s [3]. The current study reveals that the crossover between black and white women occurs at younger ages and earlier stages than previously noted [3]. This age group (< 40 years) is not affected by screening, as mammography in women under age 40 is under the discretion of the physician, and is rarely screening in nature unless there is a strong family history [14]. Other factors may play a significant role in the apparent crossover in incidence rates for white and black women.

Although several researchers have suggested a possible link between this crossover effect and socioeconomic status [3, 15], no studies have adequately examined this relationship to date [16, 17]. For breast cancer diagnosed after age 40 years, nulliparity and a small number of full-term pregnancies are two factors associated with small increases in risk indicating that full-term pregnancy occurring before age 40 confers protection from breast cancer at later ages. However, the immediate effect of a full-term pregnancy in younger women appears to be a short-term increased risk for breast cancer [16]. Black women have higher parity at all ages compared with white women [18], which may contribute to the apparent crossover effect of higher breast cancer incidence in younger black women. This finding has great public health implications, because breast cancer in young black women has been shown to be quite aggressive [19].

Another new finding from this study is the phenomenon of Clemmenson's Hook, which describes

Table 3. Average annual age-specific breast cancer incidence rates per 100,000 population stratified by disease stage for racial groups in the NAACCR data base, 1994–1998

Age (years)	White[a] Rate	Black[b]			American Indian[c]			Asian/Pacific Islander[d]		
		Rate	RR	95% CI	Rate	RR	95% CI	Rate	RR	95% CI
Localized stage										
20–24	0.59	–[g]	–	–	–	–	–	–	–	–
25–29	3.61	5.01	1.39	1.15–1.69	–	–	–	2.69	0.75	0.52–1.08
30–34	12.47	14.84	1.19	1.07–1.32	–	–	–	9.81	0.79	0.66–0.95
35–39	30.07	29.81	0.99	0.92–1.07	–	–	–	25.60	0.85	0.76–0.95
40–44	65.89	56.77	0.86	0.81–0.91	33.88	0.51	0.36–0.73	56.50	0.86	0.79–0.93
45–49	111.14	92.64	0.83	0.79–0.87	53.61	0.48	0.35–0.66	90.54	0.81	0.76–0.87
50–54	154.84	119.39	0.77	0.73–0.80	66.80	0.43	0.31–0.60	117.15	0.76	0.71–0.82
55–59	189.89	143.92	0.76	0.72–0.80	81.62	0.53	0.38–0.74	137.97	0.73	0.68–0.79
60–64	230.94	155.85	0.67	0.64–0.70	112.20	0.49	0.36–0.67	146.04	0.63	0.58–0.68
65–69	279.26	186.08	0.67	0.63–0.70	82.64	0.30	0.20–0.45	158.19	0.57	0.53–0.62
70–74	316.30	210.50	0.67	0.64–0.70	–	–	–	178.84	0.57	0.53–0.62
75–79	325.06	219.53	0.68	0.64–0.72	–	–	–	165.80	0.51	0.46–0.56
80–84	300.40	212.74	0.71	0.66–0.76	–	–	–	160.18	0.53	0.46–0.61
85+	220.16	148.46	0.67	0.61–0.73	–	–	–	102.29	0.46	0.38–0.56
Regional stage										
20–24	0.47	1.28	2.72	1.77–4.17	–[g]	–	–	0.21	0.45	0.11–1.84
25–29	3.37	5.92	1.76	1.47–2.11	–	–	–	2.43	0.72	0.49–1.05
30–34	10.74	14.65	1.36	1.22–1.52	–	–	–	7.42	0.69	0.56–0.85
35–39	23.94	30.28	1.26	1.17–1.36	–	–	–	19.10	0.80	0.70–0.91
40–44	41.82	47.65	1.14	1.07–1.21	–	–	–	35.69	0.85	0.77–0.94
45–49	66.67	71.18	1.07	1.01–1.13	45.15	0.68	0.48–0.96	53.14	0.80	0.73–0.88
50–54	80.62	85.91	1.07	1.01–1.13	46.39	0.58	0.39–0.86	57.70	0.72	0.65–0.80
55–59	89.02	93.25	1.05	0.99–1.12	58.30	0.65	0.44–0.96	69.56	0.78	0.70–0.87
60–64	95.54	89.14	0.93	0.87–0.99	74.80	0.78	0.53–1.15	68.24	0.71	0.63–0.79
65–69	101.97	103.88	1.02	0.96–1.09	–	–	–	58.68	0.58	0.51–0.66
70–74	109.30	103.77	0.95	0.88–1.02	–	–	–	50.73	0.46	0.40–0.53
75–79	108.50	110.35	1.02	0.94–1.10	–	–	–	51.84	0.48	0.40–0.57
80–84	102.44	98.62	0.96	0.87–1.07	–	–	–	38.14	0.37	0.28–0.49
85+	85.36	87.04	1.02	0.91–1.14	–	–	–	26.56	0.31	0.21–0.45
Distant stage										
20–24	–[g]	–	–	–	–	–	–	–	–	–
25–29	0.45	–	–	–	–	–	–	0.17	0.38	0.09–1.55
30–34	1.21	2.39	1.98	1.50–2.62	–	–	–	0.91	0.75	0.41–1.38
35–39	2.28	4.77	2.09	1.72–2.54	–	–	–	1.61	0.71	0.45–1.12
40–44	4.07	8.08	1.99	1.70–2.33	–	–	–	2.96	0.73	0.52–1.03
45–49	6.96	12.83	1.84	1.61–2.11	–	–	–	5.89	0.85	0.65–1.11
50–54	10.19	17.07	1.68	1.47–1.92	–	–	–	7.72	0.76	0.58–1.00
55–59	13.35	21.25	1.59	1.39–1.82	–	–	–	8.17	0.61	0.45–0.83
60–64	17.10	25.64	1.50	1.32–1.71	–	–	–	8.91	0.52	0.38–0.71
65–69	19.65	27.16	1.38	1.21–1.57	–	–	–	8.12	0.41	0.29–0.57
70–74	22.77	30.70	1.35	1.18–1.54	–	–	–	8.70	0.38	0.27–0.55
75–79	24.81	31.17	1.26	1.08–1.46	–	–	–	9.43	0.38	0.25–0.58
80–84	26.67	34.08	1.28	1.07–1.53	–	–	–	9.15	0.34	0.19–0.60
85+	22.14	31.12	1.41	1.16–1.71	–	–	–	9.84	0.44	0.24–0.82

[a] Reference group includes data from all consenting US registries.
[b] Includes data from all consenting US registries except Idaho, Montana, New Hampshire, Utah, and Wyoming.
[c] Includes data from Arizona, Montana, North Carolina, Washington, and Wisconsin.
[d] Includes data from Arizona, Los Angeles and Greater Bay Area (California), Colorado, Connecticut, Metro Atlanta (Georgia), Hawaii, Illinois, Iowa, Kentucky, Louisiana, Metro Detroit (Michigan), New Jersey, North Carolina, Rhode Island, Utah, Washington, and Wisconsin.
[e] Rate ratio comparing each race group rate with that of whites.
[f] Confidence interval.
[g] Rate could not be calculated because fewer than 20 cases were reported.

Table 4. Average annual age-specific breast cancer incidence rates per 100,000 population stratified by disease stage for Hispanic[a] and non-Hispanic[b] ethnic groups in the NAACCR data base, 1994–1998

Age	Localized stage				Regional stage				Distant stage			
	Non-Hispanic Rate	Hispanic			Non-Hispanic Rate	Hispanic			Non-Hispanic Rate	Hispanic		
		Rate	RR[c]	95% CI[d]		Rate	RR	95% CI		Rate	RR	95% CI
20–24	0.76	–[e]	–	–	0.51	–	–	–	–	–	–	–
25–29	3.84	2.55	0.66	0.47–0.93	3.50	3.55	1.01	0.75–1.36	0.42	–	–	–
30–34	12.66	7.76	0.61	0.50–0.74	10.61	7.96	0.75	0.62–0.91	1.11	1.32	1.19	0.73–1.94
35–39	30.24	17.60	0.58	0.51–0.66	24.30	17.90	0.74	0.65–0.85	2.40	1.42	0.59	0.37–0.94
40–44	65.76	40.33	0.61	0.55–0.67	42.49	33.91	0.80	0.72–0.89	4.13	4.28	1.04	0.75–1.43
45–59	113.22	69.70	0.62	0.57–0.68	69.76	51.96	0.74	0.67–0.82	7.24	5.66	0.78	0.58–1.06
50–54	163.06	96.17	0.59	0.54–0.64	86.06	53.46	0.62	0.55–0.70	10.33	7.46	0.72	0.53–0.98
55–59	196.42	117.94	0.60	0.55–0.66	95.70	57.50	0.60	0.53–0.68	13.23	8.38	0.63	0.45–0.88
60–64	238.85	133.40	0.56	0.51–0.61	99.86	66.98	0.67	0.59–0.76	16.60	13.40	0.81	0.60–1.09
65–69	283.17	153.70	0.54	0.49–0.59	99.94	68.21	0.68	0.59–0.78	18.32	15.12	0.83	0.61–1.11
70–74	323.37	179.82	0.56	0.51–0.62	108.50	70.90	0.65	0.56–0.76	20.62	7.53	0.37	0.23–0.58
75–79	331.59	174.67	0.52	0.46–0.58	113.67	75.10	0.66	0.55–0.79	22.20	11.95	0.54	0.35–0.83
80–84	313.45	144.16	0.46	0.39–0.54	102.73	63.97	0.62	0.49–0.78	23.17	11.09	0.48	0.28–0.83
85+	229.21	95.84	0.42	0.35–0.51	86.31	50.68	0.59	0.45–0.77	20.81	11.98	0.58	0.33–1.01

[a] Includes data from Los Angeles and Greater Bay Area (California), Colorado, Connecticut, Idaho, Iowa, Kentucky, Metro Detroit (Michigan), Kentucky, Nebraska, Rhode Island, Utah, Washington, and Wyoming.
[b] Reference group for rate ratios.
[c] Rate ratio comparing Hispanic with non-Hispanic rates.
[d] Confidence interval.
[e] Rate could not be calculated because fewer than 20 cases were reported.

smaller increases in breast cancer incidence rates with increasing age after the age of menopause, and was apparent in all race categories (Figure 1). Although the increase in incidence rates for API women ages 50–74 in the United States is smaller compared with increased rates for white and blacks, this increase is noteworthy because it contradicts the consistent decline in breast cancer incidence beginning at about age 50 years in Japanese women in Japan [20]. Although this study did not stratify data by specific Asian subgroup, the identification of this trend supports the hypothesis that differences in environmental factors between countries (e.g., diet, obesity, alcohol use, use of hormone replacement therapy) are associated with breast cancer incidence.

The increase in breast cancer incidence noted over the past 20 years is explained in part by greater numbers of *in situ* cases detected through screening mammography. However, widespread use of screening does not account for all of the increase, and it does not account for all variations in incidence by age and race [21]. Nevertheless, increased screening may indicate increased access to health care and preventive services. The National Health Interview Survey (NHIS) examined prevalence of mammography screening over the period recommended by the US Preventive Task Services Task Force (within the past two years) for the three major racial/ethnic groups (whites, blacks, and Hispanics) in three age groups (40–49 years, 50–64 years, and 65+ years). Breen et al. [22] reviewed trends from NHIS during 1987–1998 and found that racial disparity in breast cancer screening between black and white women existed in 1987 but disappeared in 1998. Although the percentage of women reporting recent mammograms was lower among older black women compared with older white women, differences were not statistically significant. In terms of age variations, mammography use for black women in the younger age group (40–49 years) increased by 5%, while in the older age groups (50–64 years and 65+ years) use increased for black women by 13 and 16%, respectively. Hispanic women were still behind white and black women in all age groups – 60% compared with 67% of all women. In the most recent NHIS report, women with no usual source of care, recent immigrants, and the uninsured are falling further behind in screening use [23].

The 1997 Behavioral Risk Factor Surveillance Survey (BRFSS) reported prevalence of mammography screening (within the recommended past two years) was 71% for whites, 73% for blacks, and 72% for API. However, screening prevalence for AI/AN remained behind other racial groups with 60% [24]. Blackman et al. [25] reviewed BRFSS trend data for 1989–1997 and found that (1) mammography use was consistently lower for Hispanic versus non-Hispanic women; (2) women aged 50–69 years were more likely to have received a mammogram than women in younger and older age groups; and (3) women aged 70+ were consistently less likely to have had a mammogram within the past two years. By 1997, 10-year age groups for women ages 50 and over (50–59, 60–69 and 70+) who reported mammography during the past two years were similar for black and white women. Also noteworthy in 1997 black women in the youngest age group (40–49) reported a higher screening prevalence (71%) than did white women (64%) of the same age.

A study by Brinton et al. [26] investigated possible reasons for the crossover effect. The authors suggested that known risk factors (fewer births, later age at first birth, other reproductive and menstrual factors) were responsible for higher population attributable risk (PAR) for breast cancer in older white women compared with older black women and much lower PAR in younger black women. This suggests that these risk factors may be more prevalent among older women compared with younger women. Also, because breast cancer is less aggressive in older women [2], it may suggest that, although younger black women (and possibly AI and Hispanic women) are more likely to have behaviors protective of breast cancer incidence (e.g., younger age at first birth, more births), these behaviors may lead to more aggressive breast cancer.

One Michigan study hypothesized that the crossover effect could be explained by differences in the distribution of reproductive risk factors [27]. The study focused on first full-term pregnancy (FFTP), parity, and oral contraceptive and hormone replacement therapy in younger versus older black and white populations. Based on the effects of age at FFTP and parity, the authors concluded that the crossover effect between the two races is expected. Similar results were noted by Palmer et al. [28], who concluded that increased parity was associated with increased risk in women younger than age 45 years and decreased risk in those ages 45+ years. This may help explain the crossover effect between black and white women.

A major strength of the present study is that the sample is based on a much larger geographic area than previous crossover studies, and data come from population-based registries. Selection biases found in non-population-based studies are reduced or eliminated here. This study is not only based on a larger sample, but it provides newer data and describes both age- and stage-specific differences among racial and ethnic groups. Rates of breast cancer incidence in this study are very similar to those reported in the most recent *Annual Report to the Nation on the Status of Cancer* [5]. In this report, age-adjusted breast cancer incidence rates for 1995–1999 were standardized to the 2000 US standard population and reported for women of all races by age categories (all ages, 20–49, 50–64, 65–74 and 75+). Data sources included SEER and the National Program of Cancer Registry (NPCR) areas meeting high data quality standards and covering approximately 55% of the US population. The age-adjusted breast cancer incidence rates for women of all races (131.9 per 100,000) in the *Annual Report to the Nation* agree closely with those observed in the present study (129.1 per 100,000) for 1994–1998. Because the *Annual Report to the Nation* includes the same data source as the current study, along with additional SEER and NPCR data, the similarity of results supports the generalizability of the current study's findings, particularly by racial group.

This report includes racial data for AI and API racial groups as well as Hispanic ethnicity. Earlier studies of the crossover effect were limited to black and white racial groups. While uniform standards for collecting racial and ethnic data among cancer registries do not exist, the NAACCR selection criteria ensures high quality data in general from the participating registries.

Measures of cancer incidence rates for various racial and ethnic populations may be limited by problems in ascertaining race or in differences in the quality of information (resulting from errors in reporting race and ethnicity on records used for cancer incidence and at-risk populations) [29–33]. Rates may be biased because of errors in reporting race on death certificates [30] and hospital medical records (used to determine numerators for cancer incidence rates) and on censuses and surveys (used to determine denominators). Recent evaluation studies suggest that reporting race for white and black populations is generally reliable, but that biases are more serious for some of the smaller populations, particularly for AI [31, 33]. Although inter-censal estimates of denominator data have been criticized as producing inaccurate results, a recent report noted that the effect on cancer rates is modest overall [34].

Cancer registries rely primarily on medical records for their case data. The quality of data on race and ethnicity is limited by the quality of information available in records. Misclassification bias may be significant, and numerators and denominators used to calculate rates may not be equivalent, depending on how people self-identify in the census versus at a hospital when obtaining care. The 'race/ethnicity' question appears on most admission forms, but if not completed by the patient, the question may not be asked by admission staff. AIs are most likely under-represented in the counts of cancer cases. In addition, because Hispanic ethnicity is not included in racial categorization, ethnicity may be sharply under-reported. This could explain the particularly low incidence rates for AIs and Hispanics. NAACCR is currently assessing the feasibility of applying a standard method for producing incidence rates for specific populations other than whites and blacks. Although the project was completed in April 2003, the algorithm was not available at the time this study was conducted. As of April 2003, the NAACCR Hispanic Identification Algorithm (NHIA) was considered best practice, and by December 2003 was accepted as the standard by SEER, NPCR and NAACCR for enhancing Hispanic identification [35, 36].

Comparability problems are possible when rates from racial groups in some states are compared with rates from groups in different states. To determine the extent of this limitation, analyses were conducted comparing all consenting registries with those selected because they had 20 or more cases. No difference between rates by race were found, but rates between Hispanic and non-Hispanic women were significantly different. This supports the validity of the main focus of this paper – comparing the crossover effect by race – but reinforces the limitations of results by ethnicity.

A further limitation of the present analysis is the lack of information about subgroups of women within broader racial and ethnic categories. For example, estimating breast cancer incidence rates was not possible for different Hispanic groups, such as those from Mexico, Cuba, or South or Central America. In addition, specific Asian groups, such as Japanese or Chinese women, were not analyzed separately.

Although the focus of this study was breast cancer incidence, the relationship of incidence to subsequent mortality is of great interest, especially when examining disparities in breast cancer mortality between white and black women. Mortality rates for breast cancer have decreased among white women in the last 20 years, from 33 to 26 per 100,000 but the mortality rate among black women has stabilized at approximately 36 per 100,000 [37]. Studies have noted a persistent disparity in mortality despite improvements in mammography screening prevalence and early detection among black women [38, 39]. To address this disparity, further research is needed in at least three areas: disparities in health care screening and treatment [40] targeted at women in racial/ethnic minority groups and younger ages; increased risks related to biological and hormonal patterns [41–43]; and other potentially modifiable, but relatively well-documented risk factors, including obesity, physical activity, dietary patterns, and lower socioeconomic status [44–46]. In addition, including all states in the analysis as well as data collected since 1999 would allow more updated and generalizable analyses.

In summary, the results of this study highlight racial and ethnic differences in the age-and stage-specific rates for breast cancer among US women. In particular, the age at which incidence increases for black and Hispanic women, previously identified as the crossover effect, was found to be younger than previously reported (30–34 years for Hispanic versus non-Hispanic women, and 40–44 years for black versus white women). The consistently higher rates diagnosed at the distant stage for black women of all ages compared with white women have great implications for public health planning and cancer control initiatives to reduce racial/ethnic disparities. These results emphasize the need for continued public health programming that focuses on regular mammography beginning at age 40, particularly for black women, and continuing into the oldest age groups for all women. Younger women with known genetic or familial risks are guided towards earlier screening, but the racial differential is not as well understood. Further research is warranted to determine reasons for racial and ethnic differences in breast cancer, specifically examining the interaction and impact of socioeconomic status and race/ethnicity on disparities in incidence and mortality.

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